

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Coordinate Geometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COORDINATE GEOMETRY · CH₃ · L₂

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The Map

This lesson collapses Coordinate Geometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 60 website questions, 60 checked solutions, 3 local PDFs read.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: (9, -1.5), gradient -3, and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q22

Find the midpoint

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2} \right)$$

Find the gradient

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

Finish: $j = -2, k = 11$

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = 80$ H is the point with coordinates $(-4, 1)$ J is the point with coordinates $(j, 15)$ where $j < 0$ K is the point with coordinates $(6, k)$ M is the midpoint of JK. The gradient of HM is 2 Find the value of j and the value of k. $j = k =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use midpoint
"gradient", "tangent", "rate of change"	Use gradient
"work out", "find", a missing value	Find equation of a line
"work out", "show", "hence", a familiar exam phrase	Check if a point lies on a line
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.